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Werk

Titel: Inventiones Mathematicae
Verlag: Springer
Jahr: 1992
Kollektion: Mathematica
Digitalisiert: Niedersächsische Staats- und Universitätsbibliothek Göttingen
Werk Id: PPN356556735_0107
PURL: http://resolver.sub.uni-goettingen.de/purl?PPN356556735_0107

LOG Id: LOG_0013
LOG Titel: Local L-factors of motives and regularized determinants.
LOG Typ: article

Übergeordnetes Werk

Werk Id: PPN356556735
PURL: http://resolver.sub.uni-goettingen.de/purl?PPN356556735
OPAC: http://opac.sub.uni-goettingen.de/DB=1/PPN?PPN=356556735

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Invent. math. 107, 135–150 (1992)

Inventiones mathematicae
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Local L-factors of motives and regularized determinants

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Oblatum 18-XII-1990

Introduction

For a smooth projective variety X over a global field k consider the completed L-function of $H^w(X)$. It is defined as the Euler product over all places v of k of the local L-factors $L_v(H^w(X), s)$ introduced in [Se]. In the function field case the generalized Lefschetz trace formula in l-adic cohomology gives a global cohomological expression for the L-function. This has been used on the following purposes:

- to prove meromorphic continuation (even rationality in p^{-s} where $p = \operatorname{char} k$)
- for the functional equation using Poincaré duality
- to prove assertions about the zeros and poles using the theory of weights
- to study special values.

It has not been possible to extend the success of the cohomological method to the number field case. The obvious obstacle even for a conjectural approach lies in the different natures of the local factors. They are polynomials in the various transcendental functions Nv^{-s} for $v \nmid \infty$ and products of shifted Γ-functions for $v \mid \infty$. The former are determined by the galois action on l-adic cohomology (with $v \nmid l$) the latter by the real Hodge structure on singular cohomology of $X_v = X \otimes k_v$. In this note together with [D] we introduce a new point of view. For every place v the local factor is written as a "regularized characteristic power series" on a suitable infinite dimensional $\mathbb{C}$-vector space. For the infinite places and for those finite places where $H^w(X)$ has good reduction and Frobenius at v acts semisimply on the w-th l-adic cohomology we prove:

$$L_v(H^w(X), s) = \det_\infty\left(\frac{\log N_v}{2\pi i}(s\ \mathrm{id} - \Theta_v)\,\middle|\, H^w(X_v/\mathbb{L}_v)\right)^{-1}. \tag{0.1}$$

Here $\det_\infty$ is a regularized determinant, $\log Nv := i$ for $v \mid \infty$ and $H^w(\ /\mathbb{L}_v)$ is a cohomology theory on pure good reduction motives over k_v. The coefficients $\mathbb{L}_v$ are rings of Laurent polynomials over $\mathbb{C}$ if $v \nmid \infty$ and polynomial rings over $\mathbb{C}$ if

$v|\infty$. In both cases a derivation Θ_v of $\mathbb{L}_v$ is specified and $H^w(\ /\mathbb{L}_v)$ takes values in the category of $\Delta_v = \mathbb{L}_v[\Theta_v]$-modules which are free of finite rank as $\mathbb{L}_v$-modules. Only very simple (in particular holonomic) Δ_v-modules appear. For the above formula the weaker $\mathbb{C}[\Theta_v]$-structure on $H^w(X_v/\mathbb{L}_v)$ suffices.

As in [D] the cohomologies $H^w(X_v/\mathbb{L}_v)$ for $v \nmid \infty$ are constructed from known ones by using simple analogues of Fontaine's Barsotti-Tate rings §2. The corresponding Fontaine theory [F] has been developed in §1.

In §4 we prove by using explicit formulas in analytic number theory a consequence for the Riemann zeta function which is suggested by (now) obvious speculation (§3) on an arithmetic site containing $\operatorname{Spec}\mathbb{Z} \cup \{\infty\}$.

Acknowledgements. I would like to thank N. Kurokawa and Ch. Soulé for interesting comments on the preprint version of this paper (see 3.4) and U. Jannsen for explaining to me some of his work. Part of this note was done during a stay at the Max Planck Institut in Bonn whose support is gratefully acknowledged.

1 The functors $\mathbb{D}$ and $\mathbb{H}$

(*1.1*) Consider a non-archimedian local field K whose residue field κ has $q = p^f$ elements. The Weil group W_K of K sits in an exact sequence

$$1 \to I \to W_K \xrightarrow{\| \|} q^{\mathbb{Z}} \to 0$$

where I is the inertia group of K and $\| w \|$ is the power to which w raises elements in κ. The geometric Frobenius elements Φ in W_K are characterised by $\| \Phi \| = q^{-1}$. Recall that a complex representation of the Weil-Deligne group W'_K consists of
(1) A $\mathbb{C}$-vector space H and a homomorphism

$$\rho : W_K \to \mathrm{GL}(H)$$

whose kernel contains an open subgroup of I.
(2) A nilpotent endomorphism N of H such that

$$\rho(w) N \rho(w)^{-1} = \| w \| N \qquad \text{for all } w \text{ in } W_K .$$

We set $H_N = \operatorname{Ker} N$, $H_N^I = (H_N)^{\rho(I)}$ and $H^{W'_K} = H_N \cap H^{\rho(W_K)}$. Note that $\rho(I)$ commutes with N. The representation of W'_K is called unramified if $N = 0$ and I operates trivially. For λ in $\mathbb{C}^*$ we denote by $\mathbb{C}_\lambda$ the one-dimensional unramified complex representation of W'_K on which $\rho(\Phi)$ is multiplication by λ.

For n in $\mathbb{Z}$ the n-th Tate Twist $\mathbb{C}(n)$ of the trivial representation on $\mathbb{C}$ is defined by $\mathbb{C}(n) = \mathbb{C}_{q^{-n}}$ i.e. $N = 0$ and $\rho(w)c = \| w \|^n c$ for all w in W_K and c in $\mathbb{C}$.

(*1.2*) Consider $\mathbb{C}[\mathbb{C}]$ the group ring over $\mathbb{C}$ with coefficients in $\mathbb{C}$. The typical element will be written in the form $\Sigma r_\alpha e^\alpha$ with α, r_α in $\mathbb{C}$ and a *symbol* e^α obeying the rule $e^{\alpha+\alpha'} = e^\alpha e^{\alpha'}$. For a subset $S \subset \mathbb{C}^*$ we set

$$\operatorname{Log}_q S = \{\alpha \in \mathbb{C} \mid q^\alpha \quad \text{is in } S\} .$$

(1.3) Definition (1) B is the $\mathbb{C}$-algebra $\mathbb{C}[\mathbb{C}]$ endowed with two structures:
(a) an unramified representation of W'_K such that

$$\rho(w)(e^\alpha) = \|w\|^\alpha e^\alpha \qquad \text{for all } w \text{ in } W_K .$$

(the $\rho(w)$'s are in fact $\mathbb{C}$-*algebra* isomorphisms)

(b) a $\mathbb{C}$-linear derivation Θ defined by

$$\Theta(e^\alpha) = \alpha e^\alpha .$$

(2) $\mathbb{L} = B^{W'_K} = \mathbb{C}[\mathrm{Log}_q(1)]$ the $\mathbb{C}$-algebra of Laurent polynomials in e^{α_q} where $\alpha_q = 2\pi i/\log q$. It is equipped with the derivation $\Theta = \Theta|_{\mathbb{L}}$. In Sect. 3 we also write $\mathbb{L}_q$ or $\mathbb{L}_K$ for $\mathbb{L}$.

Remark. Consider the eigenvalues of Frobenius elements in W'_K acting on the I_K-fixed modules of l-adic cohomology for $l \neq p$ of smooth projective varieties over $\bar{K}$. If it were known even in the bad reduction case that they lay in the group Z_q of q-Weil numbers then one could take $B = \mathbb{C}[\mathrm{Log}_q Z_q]$ in the following. This would be the minimal "Barsotti-Tate" ring for the theory of Sect. 2.

(1.4) We can view the elements of B as entire functions on $\mathbb{C}$ by the $\mathbb{C}$-algebra injection $B \subsetneq \mathcal{O}(\mathbb{C})$ mapping e^α to the function $z \mapsto \exp(\alpha z)$. Then Θ corresponds to ∂_z, $N = 0$ and $\rho(w)(\exp(\alpha z)) = \|w\|^\alpha \exp(\alpha z)$. To see that the map is injective one looks at the Taylor development and uses the Vandermonde determinant for example. Later on this point of view will be used to apply results from differential equations.

(1.5) For λ in $\mathbb{C}^*$ set $\mathbb{L}_\lambda = B^{\rho(\Phi) = \lambda^{-1}} = \mathbb{C}[\mathrm{Log}_q \lambda]$ a (left) $\Delta = \mathbb{L}[\Theta]$-Module which is free of rank one over $\mathbb{L}$. Clearly $\mathbb{L}_\lambda = \mathbb{L} \cdot e^\alpha$ for any α with $q^\alpha = \lambda$.

(1.6) Twists. For a Δ-module D and n in $\mathbb{Z}$ a new Δ-module $D(n)$ is defined by

$$D(n) = D \otimes_{\mathbb{L}} \mathbb{L}_{q^{-n}} .$$

This is Δ-isomorphic to D equipped with the derivation

$$\Theta_{D(n)} = \Theta_D - n\mathrm{id}$$

via the map $D \to D(n)$, $d \mapsto d \otimes e^{-n}$.

(1.7) The functor $\mathbb{D}$. We need two categories:

$\mathbf{Rep}(W'_K)$ is the Tannakian category of finite dimensional complex representations H of W'_K. The twists of H are defined by $H(n) = H \otimes \mathbb{C}(n)$. Thus $H(n) = H$ as a $\mathbb{C}$-vector space, $N_{H(n)} = N$ and $\rho_{H(n)}(w) = \|w\|^n \rho_H(w)$.

$\mathbf{Der}_K$ is the additive category of left Δ-modules D which are free of finite rank over $\mathbb{L}$. It is equipped with the tensor product $\otimes_{\mathbb{L}}$ and with duals:

$$\check{D} = \mathrm{Hom}_{\mathbb{L}}(\mathbb{D}, \mathbb{L}) \qquad \text{where } \Theta_{\check{D}}(\varphi)(d) = \Theta(\varphi(d)) - \varphi(\Theta_D(d)) .$$

Then the natural map $D \otimes_{\mathbb{L}} \check{D} \to \mathbb{L}$ is Δ-equivariant and hence a morphism in $\mathbf{Der}_K$. We define an additive functor

$$\mathbb{D} : \mathbf{Rep}(W'_K) \to \mathbf{Der}_K$$

by the formula

$$\mathbb{D}(H) = (H \otimes B)^{W'_K} = (H_N \otimes B)^{\rho(W_K)} .$$

The Δ-module structure on $\mathbb{D}(H)$ is induced by the one on B. Since

$$\mathbb{D}(H) = \bigoplus_{\lambda \in \mathbb{C}^*} H_N^\lambda \otimes \mathbb{L}_\lambda \qquad \text{where } H_N^\lambda = (H_N^I)^{\rho(\Phi) = \lambda} \tag{1.8}$$

it is free of finite rank over $\mathbb{L}$. Clearly:

$$\mathrm{rk}_{\mathbb{L}} \mathbb{D}(H) \leqq \dim H_N^I$$

with equality if and only if $\rho(\Phi)$ operates semisimply on H_N^I. A representation H is called admissible if $\mathrm{rk}_{\mathbb{L}} \mathbb{D}(H) = \dim H$ i.e. if H is an unramified W'_K-module and $\rho(\Phi)$ acts semi-simply. The full subcategory of admissible representations is denoted $\mathbf{Rep}^{\mathrm{ad}}(W'_K)$. It is a semisimple Tannakian category whose simple objects are the $\mathbb{C}_\lambda$'s with λ in $\mathbb{C}^*$. For H_1, H_2 in $\mathbf{Rep}(W'_K)$ there is a natural map of Δ-modules:

$$\mathbb{D}(H_1) \otimes_{\mathbb{L}} \mathbb{D}(H_2) \to \mathbb{D}(H_1 \otimes H_2) . \tag{1.9}$$

In particular there is a natural map

$$\mathbb{D}(H) \otimes_{\mathbb{L}} \mathbb{D}(\check{H}) \to \mathbb{D}(H \otimes \check{H}) \to \mathbb{D}(\mathbb{C}(0)) = \mathbb{L}$$

i.e. a map in $\mathbf{Der}_\kappa$

$$\mathbb{D}(\check{H}) \to \mathbb{D}(\check{H}) . \tag{1.10}$$

(1.11) Proposition. *The functor $\mathbb{D}$ commutes with twists. The map* (1.10) *is an isomorphism if H is unramified and* (1.9) *is an isomorphism if H_1 and H_2 are admissible.*

Proof. For H in $\mathbf{Rep}(W'_K)$ we have

$$\begin{aligned} \mathbb{D}(H(n)) &= \bigoplus_{\lambda \in \mathbb{C}^*} H(n)_N^\lambda \otimes \mathbb{L}_\lambda = \bigoplus_\lambda H_N^\lambda \otimes \mathbb{L}_{\lambda q^{-n}} \\ &= \mathbb{D}(H) \otimes_{\mathbb{L}} \mathbb{L}_{q^{-n}} = \mathbb{D}(H)(n) . \end{aligned}$$

If H is unramified then

$$\begin{aligned} \mathbb{D}(\check{H}) &= \bigoplus_\lambda (\check{H})_N^\lambda \otimes \mathbb{L}_\lambda = \bigoplus_\lambda \mathrm{Hom}_{\mathbb{C}}(H^{\rho(\Phi) = \lambda^{-1}}, \mathbb{C}) \otimes \mathbb{L}_\lambda \\ &= \bigoplus_\lambda \mathrm{Hom}_{\mathbb{L}}(H^{\rho(\Phi) = \lambda^{-1}} \otimes \mathbb{L}_{\lambda^{-1}}, \mathbb{L}) . \end{aligned}$$

Thus since the direct sum is actually finite

$$\mathbb{D}(\check{H}) = \mathrm{Hom}_{\mathbb{L}}(\mathbb{D}(H), \mathbb{L}) = \widetilde{\mathbb{D}(H)} .$$

For admissible H_1 and H_2 we have

$$\begin{aligned} \mathbb{D}(H_1) \otimes_{\mathbb{L}} \mathbb{D}(H_2) &= \bigoplus_\lambda \left(\bigoplus_{\lambda_1 \lambda_2 = \lambda} H_1^{\rho(\Phi) = \lambda_1} \otimes H_2^{\rho(\Phi) = \lambda_2} \right) \otimes \mathbb{L}_\lambda \\ &= \bigoplus_\lambda (H_1 \otimes H_2)^{\rho(\Phi) = \lambda} \otimes \mathbb{L}_\lambda \\ &= \mathbb{D}(H_1 \otimes H_2) \end{aligned}$$

since $\rho(\Phi)$ acts semisimply on $H_i = H_{iN}^I$ for $i = 1, 2$.

For any H in $\mathbf{Rep}(W'_K)$ there is a natural map

$$\mathbb{D}(H) \otimes_{\mathbb{L}} B \to H \otimes B \tag{1.12}$$

compatible with the Δ and W'_K-module structures on both sides. Let $H^{\mathrm{ad}} = \bigotimes_{\lambda} H_N^{\lambda}$ be the maximal admissible submodule of H.

(1.13) Proposition. *The map* (1.12) *factors naturally*

$$\mathbb{D}(H) \otimes_{\mathbb{L}} B \xrightarrow{\sim} H^{\mathrm{ad}} \otimes B \subsetneq H \otimes B .$$

In particular it is injective and an isomorphism if and only if H is admissible.

Proof. Since $\mathbb{L}_\lambda \otimes_{\mathbb{L}} B \xrightarrow{m} B$ is an isomorphism of Δ, W'_K-modules we get:

$$\begin{aligned} \mathbb{D}(H) \otimes_{\mathbb{L}} B &= \bigoplus_{\lambda} H_N^{\lambda} \otimes (\mathbb{L}_\lambda \otimes_{\mathbb{L}} B) \\ &\overset{m}{\cong} \bigoplus_{\lambda} H_N^{\lambda} \otimes B \quad \text{with diagonal action of } W'_K \\ &= H^{\mathrm{ad}} \otimes B . \end{aligned}$$

(*1.14*) *The functor* $\mathbb{H}$. We define an additive functor

$$\mathbb{H} : \mathbf{Der}_\kappa \to \mathbf{Rep}^{\mathrm{ad}}(W'_K)$$

by $\mathbb{H}(D) = (D \otimes_{\mathbb{L}} B)^{\Theta = 0}$ with W'_K-action induced by the one on B. Obviously $\mathbb{H}(D)$ is unramified and $\rho(\Phi)$ acts semisimply. Moreover

$$\dim \mathbb{H}(D) \leqq \mathrm{rk}_{\mathbb{L}} D .$$

To see this let $e_1, \ldots, e_n$ be an $\mathbb{L}$-basis of D. Then $\Theta(e_j) = \sum_i a_{ij} e_i$ with $A = (a_{ij})$ in $\mathrm{M}_n(\mathbb{L})$ and we have

$$\mathbb{H}(D) \cong \{x = (x_1, \ldots, x_n)^t \in B^n \mid \Theta(x) + Ax = 0\} .$$

Viewing elements of B as functions as in (1.4) we conclude by the elementary theory of ordinary differential equations.

According to (1.13) we have for any H in $\mathbf{Rep}(W'_K)$ natural maps of W'_K-modules:

$$\mathbb{H}\mathbb{D}(H) \xrightarrow{\sim} H^{\mathrm{ad}} \subsetneq H .$$

In particular $\mathbb{H}\mathbb{D}(H) \to H$ is an isomorphism if and only if H is admissible. For any D in $\mathbf{Der}_\kappa$ there is a natural map

$$\mathbb{H}(D) \otimes B \to D \otimes_{\mathbb{L}} B \tag{1.15}$$

compatible with the Δ, W'_K-module structures. Taking W'_K-invariants we obtain a natural map of Δ-modules

$$\mathbb{D}\mathbb{H}(D) \to D . \tag{1.16}$$

(1.17) Lemma. *The maps* (1.15) *and* (1.16) *are injective. The map* (1.15) *factors naturally*:

$$\mathbb{H}(D) \otimes B \hookrightarrow \mathbb{D}\mathbb{H}(D) \otimes_{\mathbb{L}} B \subset D \otimes_{\mathbb{L}} B$$

into an isomorphism and an inclusion.

Proof. Choosing a complex number α_λ with $p^{\alpha_\lambda} = \lambda$ for every λ *in* $\mathbb{C}^*$ we find:

$$\mathbb{H}(D) = \bigoplus_\lambda (D \otimes_{\mathbb{L}} \mathbb{L}_\lambda)^{\Theta = 0} = \bigoplus_\lambda D^{\Theta = -\alpha_\lambda} \otimes \mathbb{C} e^{\alpha_\lambda} .$$

Recall that e^{α_λ} is just a symbol! Thus:

$$\mathbb{H}(D) \otimes B = \bigoplus_\lambda D^{\Theta = -\alpha_\lambda} \otimes (\mathbb{C} e^{\alpha_\lambda} \otimes B) \overset{m}{\cong} \bigoplus_\lambda D^{\Theta = -\alpha_\lambda} \otimes B$$

with Δ acting on both factors. We get

$$\mathbb{D}\mathbb{H}(D) = \left(\bigoplus_\lambda D^{\Theta = -\alpha_\lambda} \right) \otimes \mathbb{L}$$

and hence an isomorphism

$$\mathbb{H}(D) \otimes B \xrightarrow{\sim} \mathbb{D}\mathbb{H}(D) \otimes B .$$

The non-trivial part of the argument consists in proving that the map

$$\left(\bigoplus_\lambda D^{\Theta = -\alpha_\lambda} \right) \otimes \mathbb{L} \to D$$

is injective. Assume it isn't injective.

Fix the minimal $n \geqq 1$ such that there exist for $1 \leqq v \leqq n$ elements d_v in $D^{\Theta = -\alpha_{\lambda_v}}$, $\lambda_v \in \mathbb{C}^*$ which are linearly independent over $\mathbb{C}$ in D but not over $\mathbb{L}$. Consider a non-trivial relation

$$\sum_{v=1}^{n} l_v d_v = 0 \qquad \text{with } l_v \text{ in } \mathbb{L} . \tag{1.18}$$

Since n is minimal all l_v are non-zero.

Applying Θ gives:

$$\sum_{v=1}^{n} \tilde{l}_v d_v = 0 \qquad \text{where } \tilde{l}_v = \Theta(l_v) - \alpha_{\lambda_v} l_v .$$

Thus

$$\sum_{v=1}^{n-1} (\tilde{l}_n l_v - l_n \tilde{l}_v) d_v = 0$$

and again by the minimality of n

$$\tilde{l}_n l_v = l_n \tilde{l}_v \qquad \text{for all} \quad 1 \leqq v \leqslant n-1 .$$

This gives

$$\Theta(l_n)l_v - l_n\Theta(l_v) = l_n l_v(\alpha_{\lambda_n} - \alpha_{\lambda_v}) \quad \text{for } 1 \leqq v \leqq n\ .$$

Identifying $\mathbb{L}$ with $\mathbb{C}[z, z^{-1}]$ and Θ with $rz\partial_z$ where $r = 2\pi i/\log q$ we get:

$$d\log\left(\frac{l_n}{l_v}\right) = \frac{c_v}{z} \quad \text{where } c_v = r^{-1}(\alpha_{\lambda_n} - \alpha_{\lambda_v})\ .$$

Looking at residues we see that $c_v \in \mathbb{Z}$ and hence $\alpha_{\lambda_n} - \alpha_{\lambda_v} \in 2\pi i/\log q\,\mathbb{Z}$. Thus $\lambda_n = \lambda_v$ and therefore $\alpha_{\lambda_n} = \alpha_{\lambda_v}$ i.e. $c_v = 0$. It follows that $l_v = \beta_v l_n$ for some constants β_v in $\mathbb{C}^*$. Since D is torsionfree as an $\mathbb{L}$-module the relation (1.18) implies

$$\sum_{v=1}^{n} \beta_v l_v = 0 \qquad \text{a contradiction!}$$

Let us call an object D in $\mathbf{Der}_\kappa$ admissible if

$$\dim \mathbb{H}(D) = \mathrm{rk}_{\mathbb{L}} D\ .$$

The full subcategory in $\mathbf{Der}_\kappa$ of admissible objects is denoted by $\mathbf{Der}_\kappa^{\mathrm{ad}}$. The structure of such objects will now be described:

(1.19) Proposition. *For D in $\mathbf{Der}_\kappa$ the following properties are equivalent:*

(1) *D is admissible.*
(2) *D is isomorphic to an object of the form $\mathbb{D}(H)$ for some H in $\mathbf{Rep}^{\mathrm{ad}}(W'_K)$.*
(3) *D is isomorphic to a direct sum $\bigoplus_{i=1}^{n} \mathbb{L}_{\lambda_i}$ for some λ_i in $\mathbb{C}^*$.*
(4) *The map $\mathbb{H}(D) \otimes B \subsetneq D \otimes_{\mathbb{L}} B$ is an isomorphism.*
(5) *The map $\mathbb{D}\mathbb{H}(D) \subsetneq D$ is an isomorphism.*

Proof. We have $\mathbb{D}(\mathbb{C}_\lambda) = \mathbb{L}_\lambda$ and $\mathbb{H}(\mathbb{L}_\lambda) = \mathbb{C}_\lambda$ hence (2) $\Rightarrow$ (3) $\Rightarrow$ (5) $\Rightarrow$ (2). Clearly (3) $\Rightarrow$ (1) and taking the W'_K-fixed modules on both sides also (4) $\Rightarrow$ (5). It remains to see that (1) $\Rightarrow$ (4) i.e. because of (1.17) that for admissible D the map in (4) is surjective. This follows by the same argument as in the proof of (6.11) in [D]. One has to use that $B \cap \mathcal{O}^*(\mathbb{C}) = B^*$ under the identification made in (1.4). We note this as a lemma:

(1.20) Lemma. *Assume the function f in $\mathcal{O}(\mathbb{C})$*

$$f(z) = \sum_{i=1}^{n} r_i \exp(\alpha_i z); \quad r_i, \alpha_i \quad \text{in } \mathbb{C}$$

has no zeros in $\mathbb{C}$. Then $f(z) = r\exp(\alpha z)$ for some r, α and all z.

Proof. The order of the entire function f is $\leqq 1$. Hence the genus of f is also $\leqq 1$ by a general result of Hadamard [A, Chap. 5, (3.2)]. Since f has no zeros it follows that $f = \exp g$ for some polynomial of degree $\leqq 1$.

Remark. Using (1.19) we see that any D in $\mathbf{Rep}_\kappa$ has a largest admissible subobject namely $\mathbb{D}\mathbb{H}(D)$.

We conclude this section by the following now obvious result:

(1.21) Corollary. *The functors* $\mathbb{D}$ *and* $\mathbb{H}$ *provide quasi-inverse equivalences of Tannakian categories*

$$\mathbf{Rep}^{\mathrm{ad}}(W'_K) \underset{\mathbb{H}}{\overset{\mathbb{D}}{\rightleftarrows}} \mathbf{Der}_{\kappa}^{\mathrm{ad}}$$

commuting with tensor products, twists and duals.

2 Local L-factors

For a representation H in $\mathbf{Rep}(W'_K)$ the local L-factor is defined as

$$L_K(H, s) = \det(1 - \rho(\Phi)q^{-s}|H_N^I)^{-1} .$$

We refer to [D, 1] for the notions of regularized product and determinant which will be used. The following convention concerning arguments will be understood:

(2.1) Convention. Consider a regularized product of the form $\prod_{v\in T}(z + a_v)$ where z is in $\mathbb{C}$, $T \subset \mathbb{Z}$ and (a_v) is a sequence of real numbers without a finite accumulation point. In this situation we take $\arg(z + a_v)$ in $(-\frac{3\pi}{2}, \frac{\pi}{2})$ for $|v| \gg 0$. This assigns a well defined value to the above "product". The regularized determinants in this section will be given by such regularized products so that we can apply the convention to them. Note that (2.1) is compatible with the convention in [D, (1.2)].

Note also that for $v \gg 0$ we have by our choice of argument:

$$\arg(z - v) = \arg(v - z) - \pi . \tag{2.2}$$

To me the following observation is a miracle compared to the obvious formulas at the end of Sect. 4 in [D].

(2.3) Proposition. *For complex* s *and* λ *in* $\mathbb{C}^*$ *we have*

$$\det_\infty\left(\frac{\log q}{2\pi i}(s - \Theta)\middle|\mathbb{L}_\lambda\right) = 1 - \lambda q^{-s} .$$

Proof. Fix some α_0 with $q^{\alpha_0} = \lambda$. Then:

$$\det_\infty\left(\frac{\log q}{2\pi i}(s - \Theta)\middle|\mathbb{L}_\lambda\right) = \prod_{q^\alpha = \lambda}\left(\frac{\log q}{2\pi i}(s - \alpha)\right)$$

$$= \prod_{v\in\mathbb{Z}}(z + v) \quad \text{where } z = \frac{\log q}{2\pi i}(s - \alpha_0)$$

$$= \prod_{v=0}^{\infty}(z + v)\prod_{v=1}^{\infty}(z - v)$$

$$= \left(\frac{1}{\sqrt{2\pi}}\Gamma(z)\right)^{-1}\prod_{v=1}^{\infty}(e^{-i\pi}(v - z)) \quad \text{by [D, (2.1)]}$$

$$= \left(\frac{1}{\sqrt{2\pi}}\Gamma(z)\right)^{-1} e^{-i\pi(\zeta(0,\,-z)-1)} \prod_{\nu=1}^{\infty} (\nu - z)$$

by [D, (1.1)]

$$= \left(\frac{1}{\sqrt{2\pi}}\Gamma(z)\right)^{-1} e^{-i\pi(z-\frac{1}{2})}\left(-\frac{1}{z}\right)\left(\frac{1}{\sqrt{2\pi}}\Gamma(-z)\right)^{-1}$$

by [D, (2.1) and proof]

$$= \frac{2\pi e^{-i\pi(z-\frac{1}{2})}}{\Gamma(z)\Gamma(1-z)} = 2ie^{-i\pi z}\sin \pi z = 1 - \lambda q^{-s}\,.$$

(2.4) Corollary. *For H in* $\mathbf{Rep}(W'_K)$ *we have with the above convention:*

$$\det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\,\middle|\,\mathbb{D}(H)\right)^{-1} = L_K(H^{\mathrm{ad}}, s)\,.$$

Proof.

$$\det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\,\middle|\,\mathbb{D}(H)\right) = \det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\,\middle|\,\bigoplus_\lambda H_N^\lambda \otimes \mathbb{L}_\lambda\right)$$

(a finite direct sum!)

$$= \prod_\lambda \det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\,\middle|\,\mathbb{L}_\lambda\right)^{\dim H_N^\lambda}$$

$$= \prod_\lambda (1-\lambda q^{-s})^{\dim H_N^\lambda} = L_K(H^{\mathrm{ad}}, s)^{-1}\,.$$

(2.5) *Cohomology over local fields.* Let $\mathscr{M}_K$ be the category of (pure) motives for absolute Hodge cycles over K. Fix a prime number $l \neq p$ and an embedding σ: $\mathbb{Q}_l \subsetneq \mathbb{C}$. According to [Ta, 4] the l-adic cohomology groups $H_l(M) = H^*_{\text{ét}}(M_{\bar{K}}, \mathbb{Q}_l)$ are naturally representations of W'_K over $\mathbb{Q}_l$. Thus we get a cohomology theory $H^*_{l,\sigma}$ on $\mathscr{M}_K$ with values in $\mathbf{Rep}(W'_K)$ by setting:

$$H^*_{l,\sigma}(M) = H^*_l(M) \otimes_{\mathbb{Q}_{l,\sigma}} \mathbb{C}\,.$$

Remark. It is hoped [Ta, 4] that all these theories for different l and σ are isomorphic. Note that according to conjectures of Jannsen [J2] there should also be a (logarithmic) crystalline construction ($l = p$) for it.

Let $\mathscr{M}_K^{\mathrm{ad}}$ be the full (Tannakian) subcategory of motives M in $\mathscr{M}_K$ with $H^*_{l,\sigma}(M)$ in $\mathbf{Rep}^{\mathrm{ad}}(W'_K)$. Consider the composition:

$$\mathbb{D}H^*_{l,\sigma} : \mathscr{M}_K \to \mathbf{Der}_K\,.$$

According to (1.11) it commutes with twists. On $\mathscr{M}_K^{\mathrm{ad}}$ it commutes with tensor products and duals. For M in $\mathscr{M}_K^{\mathrm{ad}}$ we therefore set:

$$H^*(M/\mathbb{L}) = \mathbb{D}H^*_{l,\sigma}(M)\,.$$

If X is a smooth projective variety over K whose absolute Hodge motive $h(X)$ is in $\mathcal{M}_K^{\mathrm{ad}}$ we set

$$H^\cdot(X/\mathbb{L}) = H^\cdot(h(X)/\mathbb{L}) .$$

On such varieties the $\mathbf{Der}_\kappa$ valued cohomology theory $H^\cdot(\ /\mathbb{L})$ has a Künneth formula and it satisfies Poincaré duality: i.e.:

$$H^n(X_1 \times X_2/\mathbb{L}) \cong \bigoplus_{p+q=n} H^p(X_1/\mathbb{L}) \otimes_{\mathbb{L}} H^q(X_2/\mathbb{L})$$

naturally and there is a natural perfect pairing:

$$H^n(X/\mathbb{L}) \times H^{2d-n}(X/\mathbb{L}) \to H^{2d}(X/\mathbb{L}) \cong \mathbb{L}(-d)$$

where $d = \dim X$.

For any motive M in $\mathcal{M}_K$ we set

$$L_K(H^w(M), s) = L_K(H^w_{l,\sigma}(M), s) .$$

According to standard conjectures this local factor is independent of the choice of l and σ. For $M = h(X)$ where X has good reduction this is a known fact. The following assertion follows from (2.4).

For M in $\mathcal{M}_K$ we have:

$$\det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\middle|\mathbb{D}H^w_{l,\sigma}(M)\right)^{-1} = L_K(H^w_{l,\sigma}(M)^{\mathrm{ad}}, s) . \tag{2.6}$$

The assertion of the next theorem is expected to hold for all w:

(2.7) Theorem. *Let X/K be a smooth projective variety. Then for $w = 0, 1$ we have*

$$\det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\middle|\mathbb{D}H^w_{l,\sigma}(X)\right)^{-1} = L_K(H^w(X), s) .$$

In particular if X has good reduction:

$$\det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\middle|H^w(X/\mathbb{L}_K)\right)^{-1} = L_K(H^w(X), s) .$$

Remark. With the notations and results of [D] the last formula also holds for $K = \mathbb{R}$ or $\mathbb{C}$ if we replace $\log q$ by i and set $H^w(X/\mathbb{L}_K) = H^w_{ar}(X) \otimes_{\mathbb{R}} \mathbb{C}$.

Proof of (2.7). In view of (2.6) we have to know that a geometric Frobenius acts semisimply on $H^w(X_{\bar{K}}, \mathbb{Q}_l)^I$ for $w = 0, 1$. For $w = 0$ this is clear since G_K acts via a finite quotient on H^0. For $w = 1$ we use the theory of the Néron model in particular [G, Proposition 2.2.5] the theory of the Raynaud scheme 6f in loc. cit., [J1, 7] and Tate's result on the semisimplicity of the Frobenius action on the Tate module of an abelian variety over a finite field. That $H^w(X_{\bar{K}}, \mathbb{Q}_l)^I$ should be a semisimple module for all w is explained in [J2].

(2.8) *Cohomology over* $\mathbb{F}_p$. Choose $l \neq p$ and $\sigma: \mathbb{Q}_l \subsetneq \mathbb{C}$. For a smooth projective variety X over $\mathbb{F}_p$ set $\mathbb{L}_p = \mathbb{L}_{\mathbb{Q}_p}$ and

$$H^{\cdot}(X/\mathbb{L}_p) = \mathbb{D}(H^{\cdot}_{l,\sigma}(X)) \quad \text{where } \mathbb{D} = \mathbb{D}_{\mathbb{Q}_p} \text{ and}$$

$$H^{\cdot}_{l,\sigma}(X) = H^{\cdot}_{\text{ét}}(X_{\bar{\mathbb{F}}_p}, \mathbb{Q}_l) \otimes_{\mathbb{Q}_l,\sigma} \mathbb{C}$$

carries the natural unramified $W'_{\mathbb{Q}_p}$-representation defined by letting $\rho(\Phi)$ act as Fr_p^*. By the Grothendieck-Serre semisimplicity conjecture (proved for $w = 0, 1$) this representation should even be admissible. For any X where this is the case we have

$$(2.9)\quad \det_\infty\left(\frac{\log q}{2\pi i}(s-\Theta)\middle| H^w(X/\mathbb{L}_p)\right)^{-1} = \det_{\mathbb{Q}_l}(1 - \mathrm{Fr}_p^* p^{-s} | H^w_{\text{ét}}(X_{\bar{\mathbb{F}}_p}, \mathbb{Q}_l))^{-1}\,.$$

It is easy to see that the functional equation of the local factor and hence also of the zeta function can be proved using the left hand formula and Poincaré duality for $H^w(X/\mathbb{L}_p)$.

3 An arithmetic site?

In this section we speculate on a global cohomological approach to L-functions of varieties over number fields which is suggested by Theorem 2.7.

There should exist a (big) site $\mathcal{T}$ containing $\overline{\mathrm{Spec}\,\mathbb{Z}} = \mathrm{Spec}\,\mathbb{Z} \cup \{\infty\}$ with the following properties:

(1) $\mathcal{T}$ is equipped with a functor $\Theta: \mathcal{T} \to \mathcal{T}$.

(2) For $X/\mathbb{F}_p$ smooth projective

$$H^w(X/\mathbb{L}_p) \cong H^w(X, \mathcal{O}_{\mathcal{T}})$$

compatible with Θ and the $\mathbb{L}_p \cong H^0(\mathrm{Spec}\,\mathbb{F}_p, \mathcal{O}_{\mathcal{T}})$ structure on both sides.

(3) For $p < \infty$ and $X/\mathbb{Q}_p$ smooth projective with good reduction $X_p/\mathbb{F}_p$

$$H^w(X/\mathbb{L}_p) \cong H^w(X, \mathcal{O}_{\mathcal{T}}) \cong H^w(X_p, \mathcal{O}_{\mathcal{T}})\,.$$

More generally for any smooth projective $X \xrightarrow{\pi} \mathrm{Spec}\,\mathbb{Q}_p$

$$\mathbb{D}H^w_{l,\sigma}(X) \cong (j_* R^w \pi_* \mathcal{O}_{\mathcal{T}})_p \quad \text{(stalk in } p)$$

where j: $\mathrm{Spec}\,\mathbb{Q}_p \subsetneq \mathrm{Spec}\,\mathbb{Z}_p$ is the inclusion. The isomorphisms should be compatible with Θ and

$$\mathbb{L}_{\mathbb{Q}_p} \cong H^0(\mathrm{Spec}\,\mathbb{Q}_p, \mathcal{O}_{\mathcal{T}}) \cong H^0(\mathrm{Spec}\,\mathbb{Z}_p, \mathcal{O}_{\mathcal{T}})\,.$$

(4) For $p = \infty$ and $X/\mathbb{R}$ smooth projective

$$H^w(X/\mathbb{L}_{\mathbb{R}}) \cong H^w(X, \mathcal{O}_{\mathcal{T}})$$

again respecting the Θ and $\mathbb{L}_{\mathbb{R}} \cong H^0(\operatorname{Spec} \mathbb{R}, \mathcal{O}_{\mathcal{F}})$ structures.

Consider now a smooth projective morphism $X \xrightarrow{\pi} \operatorname{Spec} \mathbb{Q}$ and let $j: \operatorname{Spec} \mathbb{Q} \subsetneq \overline{\operatorname{Spec} \mathbb{Z}}$ be the inclusion.

For $p \leqq \infty$ setting $\mathcal{G} := j_* R^w \pi_* \mathcal{O}_{\mathcal{F}}$ we should have

$$\mathbb{D} H^w_{l,\sigma}(X_{\mathbb{Q}_p}) \cong \mathcal{G}_p$$

and therefore with $\log \infty := i$ for $\operatorname{Re} s > \frac{w}{2} + 1$:

$$L(H^w(X), s) L_\infty(H^w(X), s) = \prod_{p \leqq \infty} \det_\infty \left(\frac{\log p}{2\pi i} (s - \Theta) \middle| \mathcal{G}_p \right)^{-1}$$

$$= \prod_{i=0}^{2} \det_\infty \left(\frac{1}{2\pi} (s - \Theta) \middle| H^i(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{G}) \right)^{(-1)^{i+1}} \tag{3.1}$$

by applying a Lefschetz trace formula. The regularized characteristic power series

$$\det_\infty \left(\frac{1}{2\pi} (s - \Theta) \middle| H^i(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{G}) \right)$$

should be entire functions in $\mathbb{C}$ and in fact polynomials for $i = 0, 2$. If $\det_\infty$ had been defined by canonical Weierstraß products it would be clear that (3.1) gave the meromorphic continuation of the L-series.

The functional equation should follow from Poincaré duality of the groups $H^i(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{G})$. Here dual should be a smooth dual as in representation theory i.e. the direct sum of the duals of the supposedly finite dimensional Θ-eigenspaces on $H^i(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{G})$. There should also be a theory of weights on these groups. It is tempting to think that the arithmetic site should also contain geometric objects underlying Kurokawa's multiple zeta functions [K3].

For the Riemann zeta function we would get from (3.1):

$$2^{-1/2} \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

$$= \frac{\det_\infty \left(\frac{1}{2\pi} (s - \Theta) \middle| H^1(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{O}_{\mathcal{F}}) \right)}{\det_\infty \left(\frac{1}{2\pi} (s - \Theta) \middle| H^0(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{O}_{\mathcal{F}}) \right) \det_\infty \left(\frac{1}{2\pi} (s - \Theta) \middle| H^2(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{O}_{\mathcal{F}}) \right)}.$$

On $H^i(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{O}_{\mathcal{F}})$ the eigenvalues of Θ should have weight i that is real part $= \frac{i}{2}$. Since the left hand side has poles only at $s = 0, 1$ and of first order, H^0 and H^2 must be onedimensional over $\mathbb{C}$ and

$$\det_\infty \left(\frac{1}{2\pi} (s - \Theta) \middle| H^i(\overline{\operatorname{Spec} \mathbb{Z}}, \mathcal{O}_{\mathcal{F}}) \right) = \begin{cases} \frac{s}{2\pi} & \text{for } i = 0 \\ \frac{s-1}{2\pi} & \text{for } i = 2 \,. \end{cases}$$

Note that $H^2 = H^0(-1)$ as $\mathbb{C}[\Theta]$-modules.

Hence we should have:

$$(3.2)\quad 2^{-1/2}(2\pi)^{-2}\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s)s(s-1) = \det_\infty\left(\frac{1}{2\pi}(s-\Theta)\,\middle|\,H^1(\overline{\operatorname{Spec}\mathbb{Z}}, \mathcal{O}_{\mathcal{F}})\right).$$

Thus the Riemann ξ-function would appear very naturally as a regularized characteristic power series on $H^1(\overline{\operatorname{Spec}\mathbb{Z}}, \mathcal{O}_{\mathcal{F}})$. The pair Θ, $H^1(\overline{\operatorname{Spec}\mathbb{Z}}, \mathcal{O}_{\mathcal{F}})$ would constitute a canonical candidate for Hilbert's well known suggestion to prove the Riemann hypotheses via the theory of selfadjoint operators.

The above formula (3.2) suggests immediately the following theorem which is proved in the next section:

(3.3) Theorem. *For* $\operatorname{Re} z > 1$ *consider the Dirichlet series:*

$$\xi(s, z) = \sum_\rho \frac{1}{\left[\frac{1}{2\pi}(z-\rho)\right]^s}$$

where ρ *runs over the non-trivial zeros of the Riemann zeta function and we take* $\arg(z-\rho)$ *in* $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. *Then* $\xi(s, z)$ *converges absolutely for* $\operatorname{Re} s > 1$ *and for fixed* z *it has an analytic continuation to a holomorphic function* $s \mapsto \xi(s, z)$ *in* $\mathbb{C} \setminus \{1\}$. *Moreover we have:*

$$(3.3.1)\quad 2^{-1/2}(2\pi)^{-2}\pi^{-\frac{z}{2}}\Gamma\left(\frac{z}{2}\right)\zeta(z)z(z-1) = \exp(-(\partial_s\xi)(0, z)).$$

(3.4) Remarks. (1) In a letter to the author Ch. Soulé [So] has extended the statement of the theorem to all z in the set Ω of complex numbers which are not of the form $\rho - \lambda$, $\lambda \geqq 0$: By a Mellin transform argument combined with a result of Cramer he proves that $\xi(s, z)$ can be continued to a meromorphic function in $\mathbb{C} \times \Omega$ regular at $s = 0$. By the unicity of analytic continuation formula (3.3.1) now extends to z in Ω. Soulé also indicates a proof of (3.3.1) which is different from the one in the next section. It uses the axiomatics of [V, C-V] to characterize both sides of the equation in terms of asymptotic behaviour.

(2) The theorem is not a formal consequence of the existence of the canonical Weierstraß product for the ξ-function. The following example was kindly pointed out to me by N. Kurokawa: The function $\prod_p(1 + \frac{z}{p})e^{-\frac{z}{p}}$ with p running over the prime numbers is entire but the function $\sum_p(p+z)^{-s}$ does not have a holomorphic continuation to $s = 0$. In fact the line $\operatorname{Re} s = 0$ is the natural boundary, c.f. [K2]. For the relation between Weierstraß products and regularized products we refer to [V, C-V].

(3) A formula which is related to (3.3.1) but closer in spirit to the theory of the Selberg trace formula was first given in [K1].

We end this section by an analogue of (3.1) for the zeta function of varieties over finite fields:

(3.5) Proposition. *Let X be smooth projective variety over $\mathbb{F}_p$ and assume that Fr_p^* acts semisimply on $H^*_{l,\sigma}(X)$. Then we have for $\mathbb{L} = \mathbb{L}_{\mathbb{Q}_p}$:*

$$\zeta_X(s) = \prod_{x \in |X|} \det_\infty \left(\frac{\log p}{2\pi i} (s \deg(x) - \Theta) \middle| \mathbb{L} \right)^{-1}$$
$$= \prod_{i=0}^{2d} \det_\infty \left(\frac{\log p}{2\pi i} (s - \Theta) \middle| H^i(X/\mathbb{L}) \right)^{(-1)^{i+1}} .$$

Proof. Using (2.3) we find:

$$\det_\infty \left(\frac{\log p}{2\pi i} (s \deg(x) - \Theta) \middle| \mathbb{L} \right) = 1 - p^{-s \deg x}$$

hence the first equation. Rewriting the usual cohomological expression for $\zeta_X(s)$ using (2.9) we conclude.

4 A representation of the ξ-function as a regularized product

In this section we give a proof of Theorem 3.3 using explicit formulas in number theory [Ba].

Fix complex numbers z, s with $\mathrm{Re}\, z$, $\mathrm{Re}\, s > 1$ and consider the function $F : \mathbb{R} \to \mathbb{C}$ defind by $F(x) = x^{s-1} \exp(-(z - \frac{1}{2})x)$ for $x \geqq 0$ and $F(x) = 0$ for $x < 0$. Then for any $0 < b < \mathrm{Re}\, z - 1$ the function $F(x) \exp((\frac{1}{2} + b)x)$ has bounded total variation on $[0, \infty)$ since it is smooth and

$$\int_0^\infty \left| F(x) \exp\left(\left(\frac{1}{2} + b \right) x \right) \right| dx < \infty .$$

For the Laplace-transform

$$\Phi(w) = \int_{-\infty}^{\infty} F(x) \exp\left(\left(w - \frac{1}{2} \right) x \right) dx$$

we find using [E, 1.5.1 (34)] if $\mathrm{Re}\, z > \mathrm{Re}\, w$

$$\Phi(w) = \Gamma(s)(z - w)^{-s} \quad \text{where} \quad -\frac{\pi}{2} < \arg(z - w) < \frac{\pi}{2} .$$

Applying explicit formulas for the Riemann zeta function to F in the form given in [Ba, Theorem 1] yields the following formula:

$$\lim_{T \to \infty} \sum_{|\mathrm{Im}\, \rho| < T} \Phi(\rho) = \Phi(0) + \Phi(1) - \sum_{p \nmid \infty} \sum_{n=1}^{\infty} \frac{\log p}{p^{n/2}} F(\log p^n) + \int_0^\infty F(x) \frac{e^{\frac{3x}{2}}}{1 - e^{2x}} dx .$$

Using formula 1.10 (3) in [E] for the Hurwitz zeta function $\zeta(s, z)$ we get:

$$(2\pi)^{-s} \xi(s, z) = \frac{1}{z^s} + \frac{1}{(z-1)^s} - \frac{1}{\Gamma(s)} \sum_{p \nmid \infty} \sum_{n=1}^{\infty} \frac{\log p}{p^{n/2}} F(\log p^n) - 2^{-s} \zeta\left(s, \frac{z}{2} \right) .$$

since $\sum_\rho |\rho|^{-1-\varepsilon} < \infty$ for any $\varepsilon > 0$.

In order to obtain the analytic continuation in s of the contribution coming from the finite primes we write it as a suitable integral transformation.

Let L_- be the contour consisting of the lower edge of the cut from $-\infty$

to $-\delta$, the circle $t = \delta e^{i\varphi}$ for $-\pi \leqq \varphi \leqq \pi$ and the upper edge of the cut from $-\delta$ to $-\infty$. According to [E, 1.6 (6)] we have for $a > 0$ the formula:

$$\frac{a^{s-1}}{\Gamma(s)} = \frac{1}{2\pi i} \int_{L_-} t^{-s} e^{ta}\, dt \ .$$

Hence for $0 < \delta < \operatorname{Re} z - 1$ we obtain:

$$\begin{aligned}\frac{1}{\Gamma(s)} \sum_{p \nmid \infty} \sum_{n=1}^{\infty} \frac{\log p}{p^{n/2}} F(\log p^n) &= \frac{1}{\Gamma(s)} \sum_{p \nmid \infty} \sum_{n=1}^{\infty} \frac{\log p}{p^{nz}} (\log p^n)^{s-1} \\ &= \frac{1}{2\pi i} \sum_{p,n} \log p \int_{L_-} p^{n(t-z)} t^{-s}\, dt \\ &= -\frac{1}{2\pi i} \int_{L_-} \frac{\zeta'}{\zeta}(z-t) t^{-s}\, dt \ .\end{aligned}$$

We can interchange the infinite sum and the integral by Lebesgue dominated convergence: observe that $\int_{-\infty}^{\delta} \frac{\zeta'}{\zeta}(\operatorname{Re}(z) - t) t^{-\operatorname{Re} s}\, dt$ converges absolutely since $\lim_{x \to \infty} \zeta(x) = 1$ and $|\zeta'(x)| \leqq C \frac{\log x}{2^x}$ for $x \geqq 2$ and some constant C. By similar arguments it follows that the above integral defines a holomorphic function of s in $\mathbb{C}$ whose derivative is obtained by differentiation under the integral. Hence for $\operatorname{Re} z > 1$ the formula

$$(2\pi)^{-s} \xi(s, z) = \frac{1}{z^s} + \frac{1}{(z-1)^s} + \frac{1}{2\pi i} \int_{L_-} \frac{\zeta'}{\zeta}(z-t) t^{-s}\, dt - 2^{-s} \zeta\left(s, \frac{z}{2}\right)$$

defines the holomorphic continuation of $s \mapsto \xi(s, z)$ to $\mathbb{C} \setminus \{1\}$. For the analytic continuation of the Hurwitz zeta function needed here we refer to [E, 1.10] as well as for the formulas

$$\zeta(0, z) = \frac{1}{2} - z \quad \text{and} \quad (\partial_s \zeta)(0, z) = \log \Gamma(z) - \frac{1}{2} \log 2\pi$$

which will be required next.

Differentiating with respect to s above and setting $s = 0$ we find

$$\begin{aligned}(\partial_s \xi)(0, z) = \xi(0, z) \log 2\pi - \log z - \log(z-1) \\ - \frac{1}{2\pi i} \int_{L_-} \frac{\zeta'}{\zeta}(z-t) \log t\, dt + \zeta\left(0, \frac{z}{2}\right) \log 2 - (\partial_s \zeta)\left(0, \frac{z}{2}\right) .\end{aligned}$$

Hence

$$\begin{aligned}(\partial_s \xi)(0, z) = \frac{1}{2} \log 2 + \frac{z}{2} \log \pi - \log \Gamma\left(\frac{z}{2}\right) - \log \frac{z}{2\pi} - \log \frac{z-1}{2\pi} \\ - \frac{1}{2\pi i} \int_{L_-} \frac{\zeta'}{\zeta}(z-t) \log\left(\frac{t}{2\pi}\right) dt\end{aligned}$$

and it remains to see that

$$\exp \frac{1}{2\pi i} \int_{L_-} \frac{\zeta'}{\zeta}(z-t) \log\left(\frac{t}{2\pi}\right) dt = \zeta(z) .$$

Letting $\delta \to 0$ and observing that $\lim_{\delta \to 0} \delta \log \delta = 0$ we get:

$$\begin{aligned}
&\frac{1}{2\pi i} \int_{L_-} \frac{\zeta'}{\zeta}(z-t) \log\left(\frac{t}{2\pi}\right) dt \\
&= \frac{1}{2\pi i} \int_{-\infty}^{0} \frac{\zeta'}{\zeta}(z-t)\left(\log\frac{|t|}{2\pi} - i\pi\right) dt + \frac{1}{2\pi i} \int_{0}^{-\infty} \frac{\zeta'}{\zeta}(z-t)\left(\log\frac{|t|}{2\pi} + i\pi\right) dt \\
&= \int_{0}^{-\infty} \frac{\zeta'}{\zeta}(z-t)\, dt = -\int_{0}^{\infty} \frac{\zeta'}{\zeta}(z+t)\, dt \\
&= \sum_{p} \sum_{n \geqq 1} \frac{\log p}{n p^{nz}} = \log \zeta(z) .
\end{aligned}$$

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